



13.3.4 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Guided Instruction and Collaboration that I knew were ...	Something I wondered during the Guided Instruction or Collaboration was ...	Something I learned during the Guided Instruction or Collaboration was ...



Unit 13, Lesson 4: Parent Functions



Benchmark

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input-output table, determine a function type that could represent it.



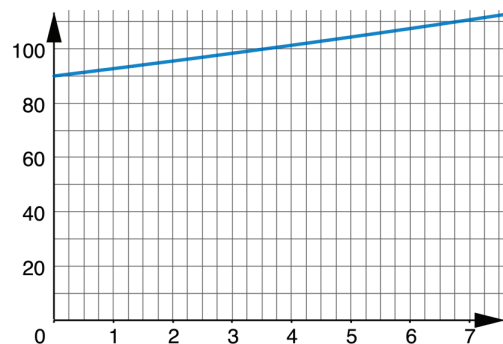
Learning Targets

- ☐ I can match multiple representations of the parent function to its table of values.
- ☐ I can determine a function type that could represent a transformation of a parent function.



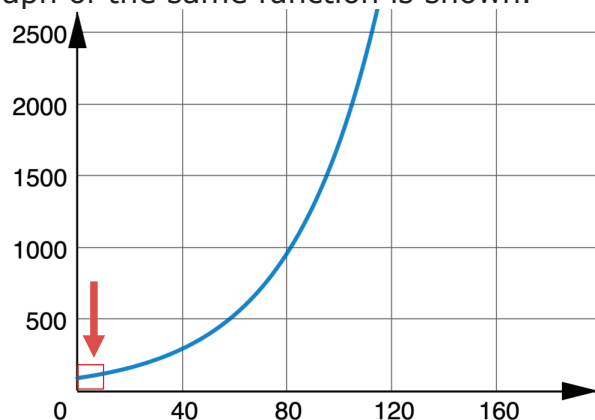
13.4.1 Warm-Up: Notice and Wonder

1. A graph of a function is shown.



What information might help you decide whether the graph represents a linear or an exponential function?

2. The graph of the same function is shown.



Explain why the x -values used for the first graph, shown in the red box, were misleading.



13.4.2 Activity: Parent Functions Card Sort

Sort the cards to match the equation, graph, and key features to the table of values of a parent function. Write the letter that corresponds to each in the table.

Table of Values	Equation	Graph	Key Features												
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-8</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>8</td></tr></table>	x	y	-2	-8	-1	-1	0	0	1	1	2	8			
x	y														
-2	-8														
-1	-1														
0	0														
1	1														
2	8														
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>2</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr></table>	x	y	-2	2	-1	1	0	0	1	1	2	2			
x	y														
-2	2														
-1	1														
0	0														
1	1														
2	2														
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr></table>	x	y	-2	-2	-1	-1	0	0	1	1	2	2			
x	y														
-2	-2														
-1	-1														
0	0														
1	1														
2	2														
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$\frac{1}{4}$</td></tr><tr><td>-1</td><td>$\frac{1}{2}$</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr></table>	x	y	-2	$\frac{1}{4}$	-1	$\frac{1}{2}$	0	1	1	2	2	4			
x	y														
-2	$\frac{1}{4}$														
-1	$\frac{1}{2}$														
0	1														
1	2														
2	4														

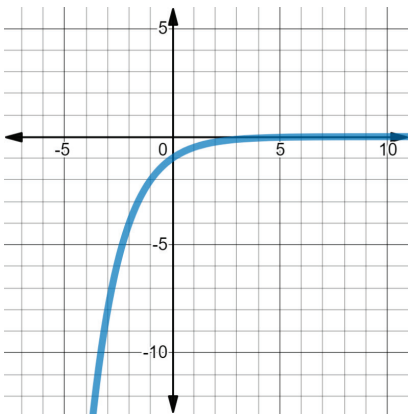
Table of Values	Equation	Graph	Key Features												
<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>4</td><td>2</td></tr><tr><td>9</td><td>3</td></tr><tr><td>16</td><td>4</td></tr></table>	x	y	0	0	1	1	4	2	9	3	16	4			
x	y														
0	0														
1	1														
4	2														
9	3														
16	4														
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>2</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>$\frac{1}{2}$</td></tr><tr><td>2</td><td>$\frac{1}{4}$</td></tr></table>	x	y	-2	4	-1	2	0	1	1	$\frac{1}{2}$	2	$\frac{1}{4}$			
x	y														
-2	4														
-1	2														
0	1														
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2	$\frac{1}{4}$														
<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4			
x	y														
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<table><tr><th>x</th><th>y</th></tr><tr><td>-8</td><td>-2</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>8</td><td>2</td></tr></table>	x	y	-8	-2	-1	-1	0	0	1	1	8	2			
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1	1														
8	2														

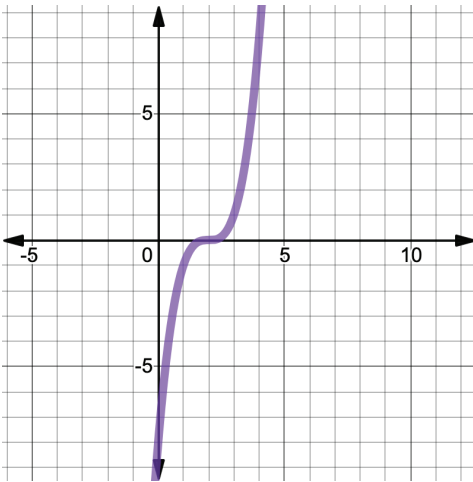


13.4.3 Exploration: Transformations of Parent Functions

Knowing the basic shape of a parent function helps to determine a transformation of the parent function.

Shown in the table are representations of transformations. Use the information from the card sort to help determine which family of functions each transformation is part of.

Representation of Transformation	Equation of Parent Function												
	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$ ☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
<table border="1"> <thead> <tr><th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>-2</td><td>3</td></tr> <tr><td>-1</td><td>4</td></tr> <tr><td>0</td><td>5</td></tr> <tr><td>1</td><td>6</td></tr> <tr><td>2</td><td>7</td></tr> </tbody> </table>	x	y	-2	3	-1	4	0	5	1	6	2	7	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$ ☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y												
-2	3												
-1	4												
0	5												
1	6												
2	7												

Representation of Transformation	Equation of Parent Function												
$y = x + 5 $	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$ ☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
<table><tr><td>x</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>$\frac{1}{4}$</td><td>$\frac{1}{2}$</td><td>1</td><td>2</td><td>4</td></tr></table>	x	-1	0	1	2	3	y	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$ ☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	-1	0	1	2	3								
y	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4								
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Representation of Transformation	Equation of Parent Function												
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x	y												
-2	6												
-1	3												
0	0												
1	3												
2	6												
$y = \left(\frac{1}{2}\right)^x - 5$	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$ ☐ $y = x $ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												



13.4.4 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about each learning target.

1. I can match multiple representations of the parent function to its table of values.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



2. I can determine a function type that could represent a transformation of a parent function.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



Unit 13, Lesson 5: Exploring Transformations – Horizontal Shifts



Benchmark

MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .



Learning Targets

- ☐ I can discover the effect on the graph of $f(x)$ for a transformation of $f(x + k)$.
- ☐ I can discover the effect on a table of values of $f(x)$ for a transformation $f(x + k)$.